EXP-4:Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

OBJECTIVES:

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

SOFTWARE / HARDWARE REQUIRED:

OptiSystem Software

Reflection:

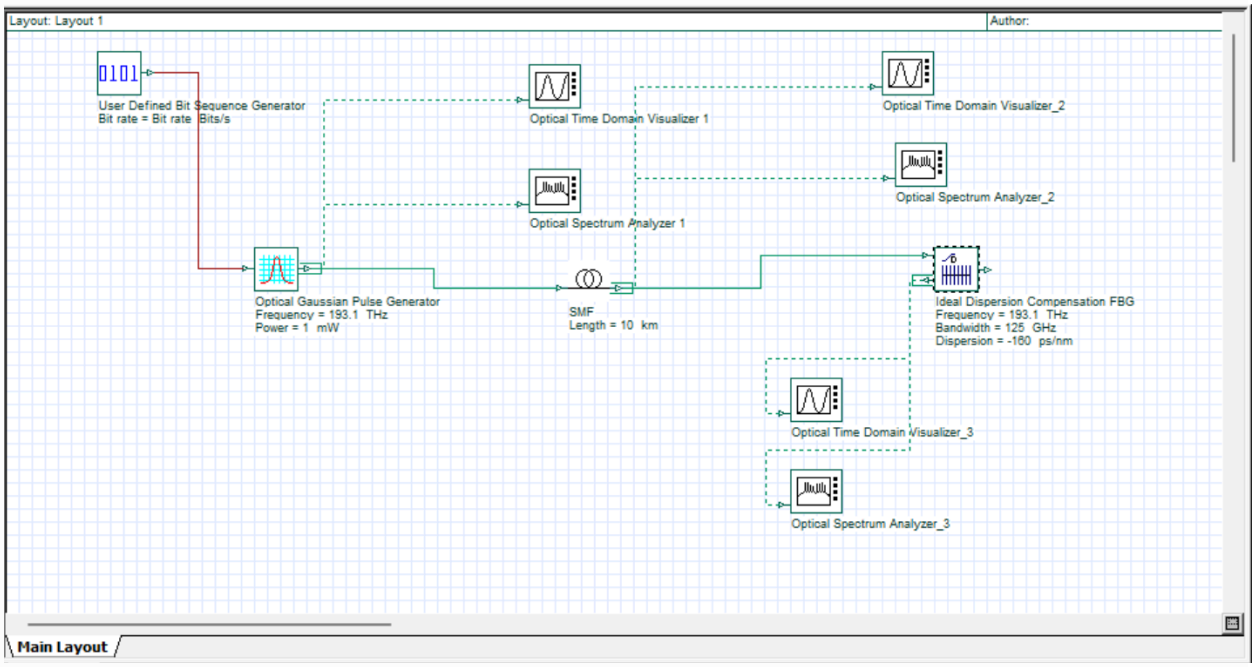
This experiment helped me understand how dispersion causes optical pulse broadening during fiber transmission.

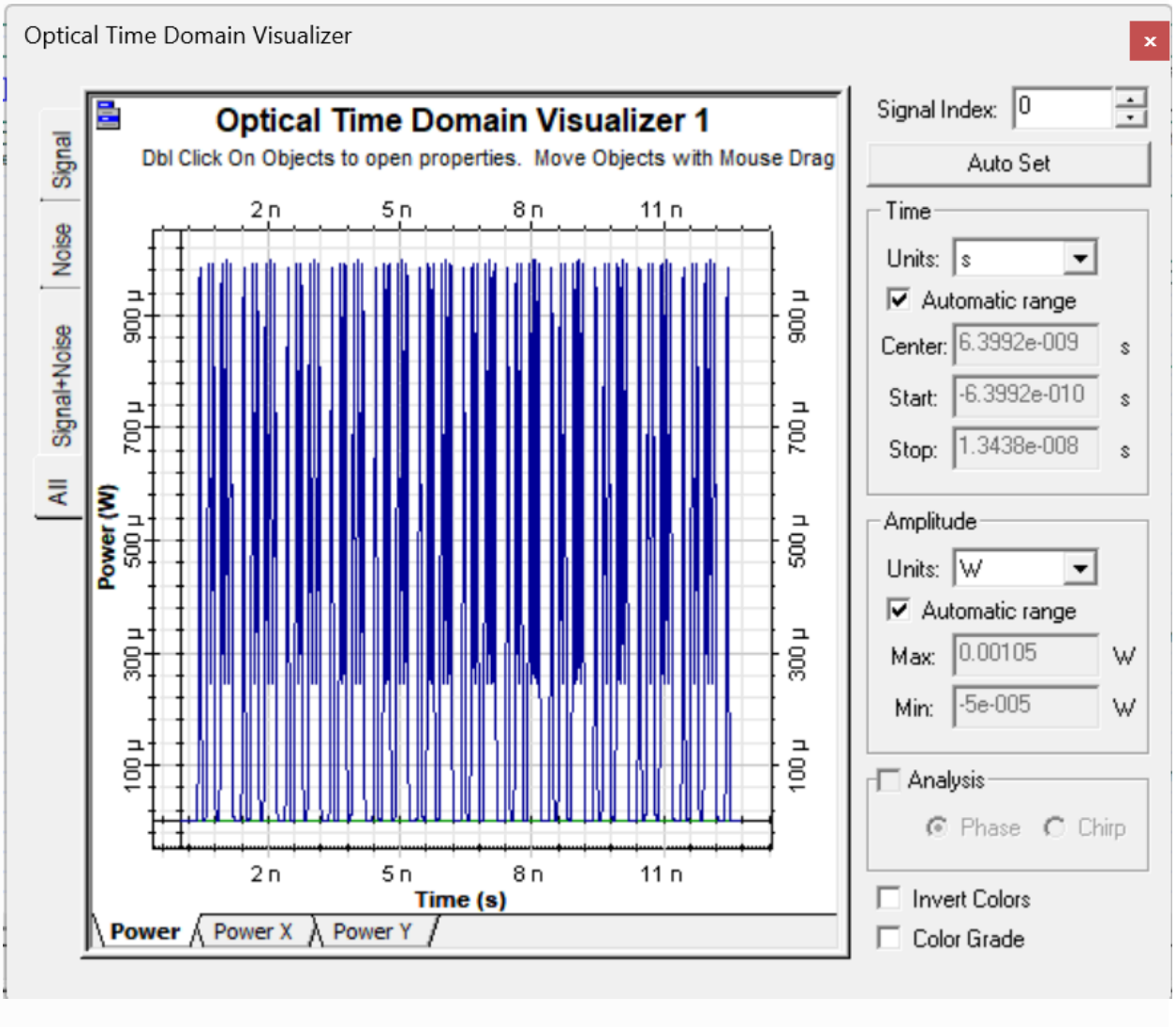
I learned that setting the compensator to −160 ps/nm can effectively restore the original pulse shape.

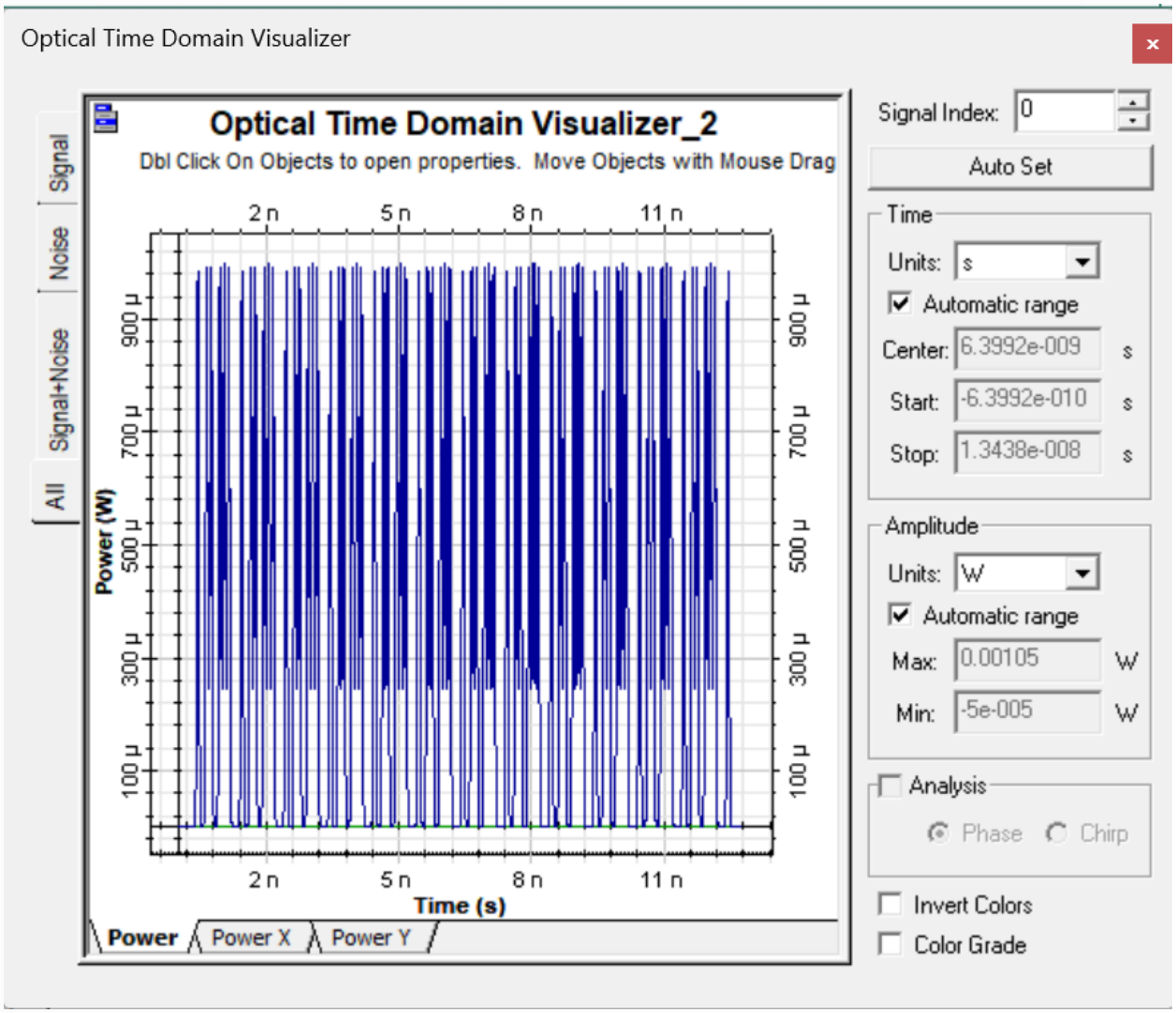
Experiment Procedure:

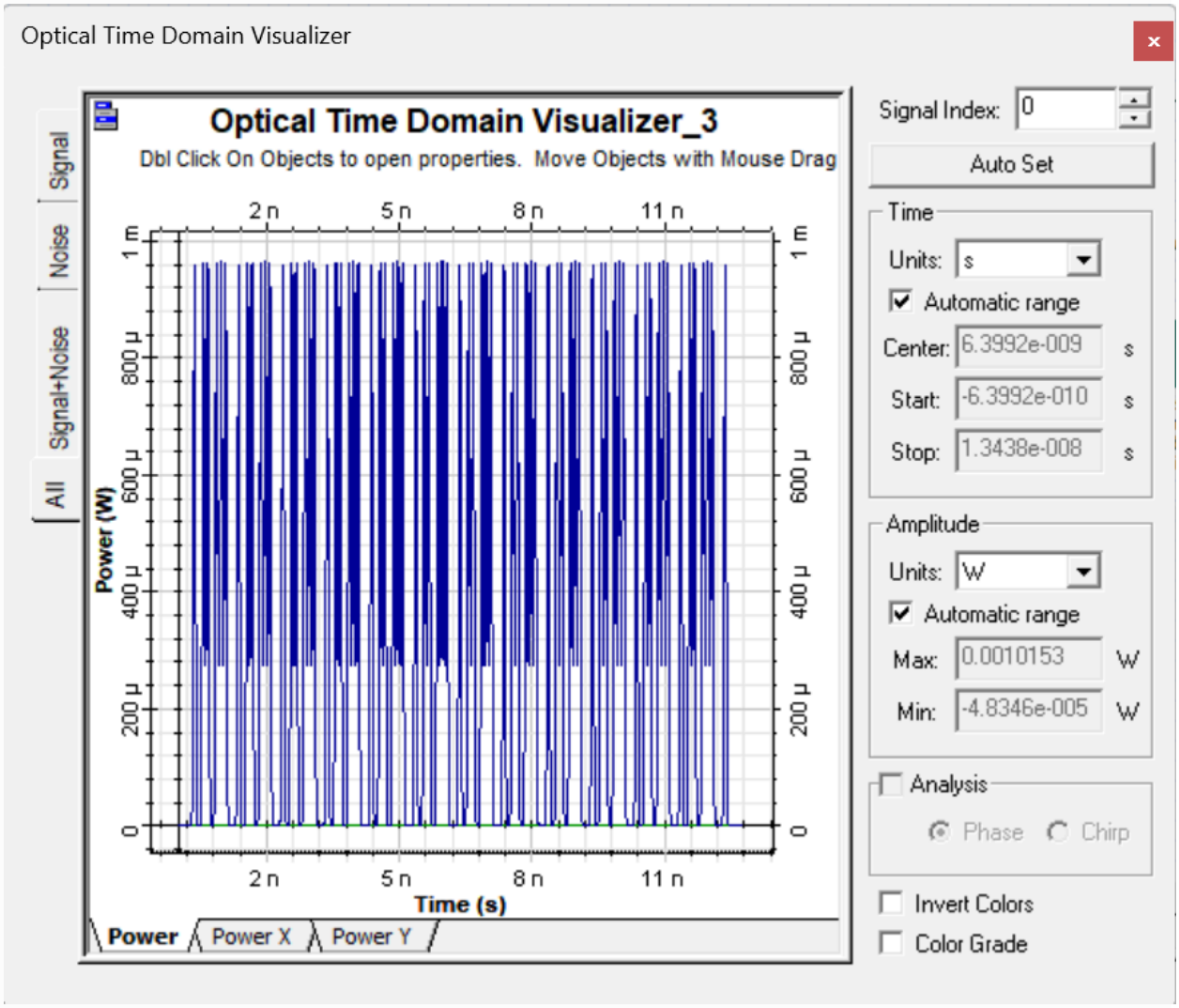
1. Open a new OptiSystem project and set the bit rate to 40 Gb/s.
2. Add the User Defined Bit Sequence Generator and Optical Gaussian Pulse Generator.
3. Set pulse frequency to 193.1 THz, power to 1 mW, and pulse width to 12.5 ps.
4. Connect a Time Domain Visualizer and Spectrum Analyzer to observe the input pulse.
5. Connect the pulse generator to SMF and set the fiber length to 10 km.
6. Observe the broadened pulse at the SMF output.
7. Connect the SMF to the Ideal Dispersion Compensation FBG and set dispersion to −160 ps/nm.
8. Observe the pulse after compensation using the visualizer and analyzer.
9. Run the simulation and compare the pulse widths before and after compensation.

OUTPUT:


Layout: Layout 1
Author:
0101
User Defined Bit Sequence Generator
Bit rate = Bit rate Bits/s
Optical Time Domain Visualizer 1
Optical Time Domain Visualizer_2
Optical Spectrum Analyzer 1
Optical Spectrum Analyzer_2
Optical Gaussian Pulse Generator
Frequency = 193.1 THz
Power = 1 mW
SMF
Length = 10 km
Ideal Dispersion Compensation FBG
Frequency = 193.1 THz
Bandwidth = 125 GHz
Dispersion = -160 ps/nm
Optical Time Domain Visualizer_3
Optical Spectrum Analyzer_3
Main Layout


Optical Time Domain Visualizer
Optical Time Domain Visualizer 1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
2 n
5 n
8 n
11 n
900 µ
700 µ
500 µ
300 µ
100 µ
Power (W)
Time (s)
Power
Power X
Power Y
Signal Index:
0
Auto Set
Time
Units:
s
Automatic range
Center:
6.3992e-009
s
Start:
-6.3992e-010
s
Stop:
1.3438e-008
s
Amplitude
Units:
W
Automatic range
Max:
0.00105
W
Min:
-5e-005
W
Analysis
Phase
Chirp
Invert Colors
Color Grade


Optical Time Domain Visualizer
Optical Time Domain Visualizer_2
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
2 n
5 n
8 n
11 n
900 µ
700 µ
500 µ
300 µ
100 µ
Power (W)
Time (s)
Power
Power X
Power Y
Signal Index:
0
Auto Set
Time
Units:
s
Automatic range
Center:
6.3992e-009
Start:
-6.3992e-010
Stop:
1.3438e-008
Amplitude
Units:
W
Automatic range
Max:
0.00105
Min:
-5e-005
Analysis
Phase
Chirp
Invert Colors
Color Grade


Optical Time Domain Visualizer
Optical Time Domain Visualizer_3
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
2 n
5 n
8 n
11 n
1 m
800 µ
600 µ
400 µ
200 µ
0
Power (W)
Time (s)
Power
Power X
Power Y
Signal Index:
0
Auto Set
Time
Units:
s
Automatic range
Center:
6.3992e-009
s
Start:
-6.3992e-010
s
Stop:
1.3438e-008
s
Amplitude
Units:
W
Automatic range
Max:
0.0010153
W
Min:
-4.8346e-005
W
Analysis
Phase
Chirp
Invert Colors
Color Grade

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, –160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of settings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

CONCLUSION: Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and setting the compensator dispersion to –160 ps/nm restored the pulse, achieving an exact compensation of the accumulated dispersion.